Gains Hide Failures: A Ledger Evaluation of a Commitment Guard for AAC Sentence Expansion

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Abstract

People who use AAC (augmentative and alternative communication) pick a few words, and a language model turns them into a sentence the app speaks, sometimes dropping a "not". We evaluate a commitment *guard*, which makes the model rewrite when its sentence contradicts the user's input, in a *ledger* that sets each gain beside what it leaves broken, what it breaks and what it costs, against free baselines. On 78 fresh speech-act families the guard raises the pass rate (joint pass) by +9.83 points [8.39, 11.32] (per model, +0.85 to +19.02) and repairs 52.51% of strict-check failures, with 0 strict-check pass-to-fail changes among 1,394 correct texts. In a human audit of 120 texts (rule frozen in advance), repairs keep the meaning that their own failed originals lost and remove invented stance; on the frozen usable endpoint they show no detectable cost against never-failed controls (71.7% against 85.0%; deployment clause unresolved), while the stricter, uncorrected endpoints do show one. Models commit to the affirmative reading in 99.9% of committed answers even when told not to assume, so their rare errors are silent reversals; supplying the missing cue, as a chip or a plain word, restores the value (silent polarity reversals per 1,000 messages, regeneration on the same items: 23.8 to 1.6 with a chip, 2.6 with a word). Writers accept 1 of 200 outputs as written, with no edit, so the ledger reports whole-message fidelity beside the commitments, and it yields post hoc tests of earlier predictions and a pre-specified human-effect design.

1 Introduction

People who use AAC often compose by selecting a few words or symbols, because every selection costs time and effort (Valencia et al. 2023; Cai et al. 2024). LLM keyword expansion turns `want go party` into a full sentence (Shen et al. 2022; Cai et al. 2024), which a system may speak on the user's behalf, sometimes after review (Hancock, Naaman, and Levy 2020; Valencia et al. 2020). But the model decides more than words: whether something happened or did not, whether the sentence asks or tells, whether it says "please". Given `not like medication`, a sentence that drops the negation, turns it into a question or adds politeness is fluent and wrong. Three kinds of fix are being explored: check the output against the user's words and regenerate on a conflict (Madaan et al. 2023; Gou et al. 2024), offer suggestions that span the readings the words leave open, or give the user a control over stance or context (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025), which our one-tap cue instantiates. Such work commonly reports gains of the accuracy or similarity kind (Cai et al. 2022; Shen et al. 2022; Cai et al. 2024), which are real. A gain-only report hides what a fix leaves unfixed, what it changes that was already right, and whether the repaired text is something a person could use.

User input	Model wrote	Guard output	What the scorer and the person see
`[?] [I] bring friend`	"Is it okay if I bring a friend?"	"Is [I] bring a friend?"	Scored a pass (it keeps the question label). Not a sentence.
`knock come`	"Please knock before you come in."	"Don't knock before you come in."	A menu meant to widen choices turned a polite request into a prohibition.
`[ASK] [YOU] [PLAIN] wear headphones`	"You wear headphones."	"Ask you to wear headphones."	Scored a pass. Not what was meant.

Figure 1: Illustrative failures found by the audit, real stored outputs. Bracketed tokens in an input are cue chips, single taps that fix one commitment: `[NOT]` negation, `[?]` a question, `[ASK]` and `[YOU]` an ask addressed to the listener, `[I]` the speaker as agent, `[PLAIN]` no added politeness. They are not typical: each is the first case in a fixed hash order within its category, except row 3, the second repair in that order; the rate of its category is given here. Row 1: the guard changed 26 of 1,394 already-correct confirmation texts, none into a strict-check failure. Row 2: menu rewrite; sets offering both readings fell from 26/40 to 12/40. Row 3: 4 of the 63 scored development repairs begin "Ask you" (post hoc lexical count).

Thesis and ledger. Where only gains are reported, failures stay hidden, so we report a ledger. Beside each benefit it records *offline failure indicators* (events in stored text that a

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gain-only report omits: failures that remain, correct outputs the fix changed, readings a menu no longer offers, silent reversals under a default that is right on average), the cost, and where the intervention applies. None of these indicators is a measured harm to a person. **Does it work?** Yes, for what it was built to protect: the guard repairs supplied-value failures on fresh items, against their own failed originals its repairs keep the meaning and remove invented stance (frozen rule: no detectable cost; deployment clause unresolved; stricter endpoints show a cost), and a supplied cue removes nearly all silent reversals; the open column is the whole message, which the ledger now measures. Our running case is keyword-to-sentence expansion with four LLMs on minimal-pair item families, defined in Section 3.

Contributions. (1) A ledger and its design rules: attribution decides what an intervention may be credited with; a free baseline and pass lines fixed in advance decide whether it merits a human study (§5). (2) One lead case, a commitment guard, evaluated end to end: on a confirmation set with a frozen rule it raises the joint pass rate by +9.83 points, repairs 52.51% of strict-check failures with 0 strict-check pass-to-fail changes (26 of 1,394 already-correct texts were rewritten), and its feedback beats plain retrying by +6.94 points [5.34, 8.60] at matched decoding (+6.20 against resampling at temperature 0.7); the ledger adds residual failures, the share a free baseline earns, and a 120-text human audit of its repairs (§6). (3) Two more ledger rows (a menu and a cue), realistic-input checks, and an analysis of the affirmative default as a good guess with an asymmetric cost. (4) Three columns to report beside commitment retention: writer-intent match, clinically weighted failure, and the measured price of a missing cue (§6.5, §6.6). (5) A pre-specified human-effect design that turns the ledger into falsifiable predictions about people, with the design implications of two AAC-user consultants (§7). We ran no study with AAC users; every result is a change in stored text.

2 Related Work

LLMs in AAC. LLM expansion and suggestion systems for AAC include KWickChat (Shen et al. 2022) and SpeakFaster (Cai et al. 2024), and AAC users have described how such systems help and hinder them (Valencia et al. 2023). Several add user-side control over stance or context (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025), one keeps a short list of bracketed function words, including `not`, in the abbreviation (Lacosse, Tano, and McNamee 2026); others measure whether affect or intent survives generation (Choudhury, Kumar, and Martin 2025). Frisch et al. argue that AI-powered AAC needs technical, user-mediated and functional evaluation together (Frisch et al. 2026); our ledger is a technical layer shaped so that a functional study can test it.

Editing, preservation and evaluation. Generate–critique–revise loops (Madaan et al. 2023; Gou et al. 2024) and conditioning a rewrite on the source dialog act (Qu, Ungar, and Sedoc 2023) are established, as is reporting a targeted change together with collateral change (Ziegenbein, Stahl, and Wachsmuth 2026; Elgaar and Amiri 2025). Preservation metrics are valid only when test data do not confound the intended change with content errors (Pauli, Augenstein, and Assent 2025); models have default politeness strategies (Zhao and Hawkins 2025), and write sentences wrongly positive about negative commonsense facts (Chen et al. 2023), so an affirmative default is not our finding; we price it. We claim neither the loop nor gain–harm accounting. We add accounting conditioned on what the input *supplied*, against a free baseline, plus a human usability check on repaired text. Suggested text shifts what people write and believe (Arnold, Chauncey, and Gajos 2020; Jakesch et al. 2023; Hohenstein et al. 2023; Agarwal, Naaman, and Vashistha 2025); we screen interventions meant to *prevent* such shifts. LLM judges (Zheng et al. 2023) can agree with each other more than with people on pragmatic labels (Wang, Sun, and Guo 2026; Panickssery, Bowman, and Feng 2024), which bounds what our machine labels can show.

3 Setting

Items and populations. An item is a minimal-pair family ("I don't like the new medication" / "I like the new medication"); each source sentence is shown to the model under five input conditions (Table 1). The *development set* has 58 constructed families on six dimensions (after quarantining two families whose scoring rule has a known blind spot); the guard was built and tested there, so its numbers are development results, and the menu and cue rows use it. The *confirmation set* has 240 fresh constructed families (polarity, speech act, politeness), 238 analyzed after a pre-generation exclusion, with three contrasts pre-specified in version control; a hash of the protocol was deposited publicly after the confirmation freeze, so it does not certify the freeze. 48 cached requests predate the final item freeze, which qualifies it. Its 78 speech-act families are the separate population on which the guard was rerun unchanged. The *human-written pairs* are 547 kept minimal pairs written by 20 non-author annotators. The *keyword study* has 20 annotators type keywords for 600 AACText messages (crowd-written fiction, not AAC users' messages).

Scoring and attribution. Four primary models write each sentence: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it (4-bit, llama.cpp) and one commercial API model; Section 6.4 adds Qwen2.5-14B and Mistral-Nemo-12B (six models from five families) and a lite Gemini model on the legend arms only (seven in all); the models are fixed, not sampled. An LLM extractor (two prompt variants) labels each output's dimensions without seeing the source, model or condition; a second-family arbiter resolves disagreements on critical labels; deterministic code turns labels into verdicts, identically for treated and untreated outputs. A target is one (item, model, input condition) outcome. Because the input condition fixes which values the model was given, a miss is attributed to one of three causes: *loss* (the input never carried the value), *violation* (it carried the value and the output changed it) or *assumption* (the input was ambiguous and the output committed anyway). Take the source "I don't like the new medication". The keywords `like medication` never carried the negation, so an af-

Glossary			
Family	Two source sentences differing on one critical dimension, with the same bare keywords	*Cue, chip*	A commitment the user adds, as one tap such as `[NOT]`, `[?]`, `[PLAIN]`, or as a plain word
Critical dimension	Polarity, agent, certainty, speech act (ask, tell, request), politeness, time/quantity	*Input conditions*	Keywords only; keywords plus a length-matched neutral word; plus the critical cue; plus all cues; the typed sentence ("typed input", the full channel)
Pass (joint pass)	Keeps the critical value and every other scored supplied predicate; the headline metric	*First pass*	The model's single greedy sentence, before any intervention
Strict-check pass	Keeps the critical value, contradicts no supplied dimension, adds no politeness; a failure is a *strict-check failure*, and *repair rate* is the share of these that the intervention fixes	*Strip-only*	Free baseline: delete an unrequested "please", "pls" or "plz" (no model call)
Violation	The input carried a value and the output changed it	*Offline failure indicator*	Event in stored text that a gain-only report omits

Table 1: Glossary. All scores are partial predicates computed from machine-extracted labels, not judgments of usability.

firmative output is a *loss* at input, and the remedy is a cue; `[NOT] like medication` carried it, so an affirmative output is a *violation*, and only then can a checker enforce the value; when the words leave both readings open and a menu of suggestions offers only one, the miss is an *assumption*, and the remedy is a menu or a question. The guard is therefore credited with violations alone, and a gain is never credited to an intervention for a failure of a kind it cannot touch. In the confirmation set, 259 disagreement rows (177 units) were resolved by a single arbiter after outcomes were known, blind to reference, model and condition; masking them as unknown gives the same three conclusions. Against two human annotators on 163 sampled untreated outputs, extractor labels agree at held-out $\kappa = 0.91$–0.97 on all 7 label fields (pre-specified gate 0.6; per-field n is smaller because ties and unsure items are excluded). In a check of arbitrated labels by 20 annotators (units where the two judges disagreed), people sided with the arbiter in 77% of codings ($\kappa = .74$) against 60% and 40% for the two judges, and 98% coded "Can you. . . ?" as a request. Repaired text is checked by the usability audit (Section 6.2). An anonymous concurrent submission (Anonymous 2027) describes the item construction and scorer and shares data with this paper; the guard and ledger results are reported here in full.

4 Interventions

The commitment guard (lead; system side). After the first pass, a deterministic lexical monitor (regexes and lexicons for negation, person, modality, question form, numbers, tense and "please") compares the output with the values in the visible input: the cues, or for typed input the monitor's own labels of the typed sentence. It also flags a politeness marker that was not supplied. On a flag, the same conversation receives one feedback turn ("Your sentence must keep: negated; no added politeness. Write the sentence again with the same meaning otherwise.") and is regenerated, up to two retries. The guard returns the first unflagged attempt, else the attempt with fewest flags, ties going to the original. The free baselines are strip-only (one failure type) and plain retrying. The no-feedback baseline (G1r) keeps the monitor, trigger and retry cap but regenerates from the original prompt with no feedback turn, at temperature 0.7 because a temperature-0 retry repeats the first pass; the guard at the same temperature (G1t) isolates the feedback message.

The menu (secondary; keywords only). From a baseline menu of three suggestions, the complete menu keeps the top one and rewrites the third (then second) slot into a missing polarity (then certainty) reading, with one model call and one monitored retry.

The cue (secondary; user side). The user adds one commitment cue, such as `[NOT]`, `[MAYBE]`, `[?]` or `[PLAIN]`. The chip is a *probe* of whether supplied information is used, not a proposed interface; the lever is the missing cue, which a plain word can also supply (Sec. 6.4).

5 Evaluation Design

Evaluation card: intervention, comparison, pass line. Pass lines were fixed in advance (development: in code before the run; confirmation: in a hashed file before any retry); verdicts are in Table 2, Fig. 2 and Section 6.2. *Guard* (targets a violation of a supplied value; compared with the stored first pass and with strip-only at zero calls): repair $\geq$30% of failures *and* change $\leq$10% of correct outputs; confirmation pass-rate gain with lower bound > 0; deployment clause, usable $\geq$ control -10 points. *Menu* (an assumption under ambiguous input; compared with the stored baseline menu): gain $\geq$30 points. *Cue* (a loss of a value at input; compared with a length-matched neutral word): gain $\geq$10 points *and* $\geq$80% of typed-input rate.

Estimand, arms and what the ledger predicts. The human target (Section 7) is the excursion effect, on a proximal "said what I meant" outcome and on weekly longitudinal rates, of delivering a regenerated sentence instead of the original at a flagged decision point; the arms are silent regeneration, flag-and-confirm and the original. The offline columns are the predictors we test against it: *changed-correct* should lower the proximal match rate, *coverage loss* (readings a menu no longer offers) the longitudinal rates, and

reliance (acceptance of flagged text rising over weeks) is a longitudinal outcome that offline text cannot show.

Paired comparisons of stored configurations. For system-side interventions, the untreated outcome is the *stored* first-pass output for the same request, and the treated outcome is what the intervention returns from it. This removes between-request variation but does not identify an effect on people: every "gain", "increment" or "effect" below is a descriptive contrast on fixed items, with no estimand for people (that estimand is the one above). Pairing matters because the same prompt does not give the same text twice (76.4–90.8% reproduced locally, 44.4–71.4% by API). Quantities are row-weighted means over targets. Intervals are 95% percentile intervals from whole-family or cluster resamples (fixed seeds), pointwise and uncorrected for selection or label error; the guard's 63/80 development repairs come from 26 families.

Offline failure indicators and cost. *Changed-correct* counts already-correct outputs whose text the intervention changed; *correct→failure* is the share that became scored failures; *residual* failures remain after the intervention. We also show what a label cannot see (scored passes that are fragments; a human usability rating of repaired text). Cost is extra model calls and latency; pass lines are coarse screens for a human study.

6 Results

6.1 The guard: about half of failures repaired, and feedback beats plain retrying

On the development cue conditions, strict-check pass rose from 94.25% to 98.78% of 1,392 targets (Table 2). Of the 80 initial failures the guard repaired 63 (78.8% [68.4, 88.9]), far above its 30% pass line. Deleting an unrequested "please" at zero calls already reaches 1,346 strict-check passes, so regeneration's increment is 2.08 points [1.15, 3.16]: 31 targets pass only with the guard and 2 only with deletion. 17 of 80 failures remained, 12 of them involving speech act, and 14/17 were still flagged after both retries: detected, not repaired. Over all development inputs (cue and no-cue conditions, so more than the 1,392 cue targets) the guard changed 49/2,154 already-correct outputs, none into a strict-check failure. Regenerating without feedback repairs little: 15/80 development failures (18.8%) against 63/80 with the guard, and 87/478 confirmation failures (18.2%) against 251/478. The guard's increment over resampling (unmatched decoding) is +3.23 [1.87, 4.81] points in development and +6.20 [4.75, 7.64] in confirmation; the matched-decoding contrast (G1t − G1r) is the clean feedback contrast: +2.95 and +6.94 [5.34, 8.60]. A retry often reproduced the first text (111/170 and 351/508 flagged requests).

Confirmation. Speech act was the guard's weakest dimension, so we reran the guard unchanged on the confirmation set's speech-act families (78 families, four models, three cue conditions, 1,872 targets). Criteria were frozen before any retry: at least 30% of strict failures repaired, at most 10% of strict-correct texts changed, and a joint-pass gain whose lower 95% bound exceeds zero. The pass rate rose from 78.79% to 88.62%, 9.83 points [8.39, 11.32]; the guard repaired 251/478 strict-check failures (52.51% [48.36, 56.70]), changed 26/1,394 already-correct texts and turned 0 of the strict-check passes into failures. Every scored criterion passed within the call caps. The gain is uneven across the four fixed models: +0.85 (API model), +3.21 (Qwen), +16.24 (Llama) and +19.02 points (Gemma), 468 targets each, so the pooled figure comes mostly from the two weaker models. The labels behind every treated-output number were validated on untreated outputs only; no human relabelled guard-treated text beyond the 60 repaired texts of the audit below. The repair rate fell from 78.8% in development to 52.51% on fresh items, so the development figure overstates it. Deletion alone gained 3.47 points, so the guard's increment over it is 6.36 points [5.29, 7.48], and 12.13% of targets remain strict-check failures. Of the results here, the guard's three frozen confirmation criteria are confirmatory; the resample baseline, the audit's secondary endpoints, H1 and H2, the decomposition and the frontier rows are post hoc or exploratory.

6.2 What the pass does not show: repair usability

A pass checks a few labels, not a sentence. For `[ASK] [YOU] [PLAIN] wear headphones`, the guard turns "You wear headphones." into "Ask you to wear headphones.", which passes every scored predicate (Figure 1). We therefore had people judge repaired text. **The audit.** A 120-text audit of new texts (60 repaired and 60 never-failed controls, plus 20 original-failed and 20 strip-only texts to separate repair from item difficulty) gave each text 3 independent non-author raters, under a rule frozen before any return: a pooled interval entirely below −10 points kills the guard for deployment (kept as a detector); one including −10 but excluding 0 is a cost of uncertain size; one including 0 is no detectable cost. Repaired texts were usable in 43/60 texts (71.7%) against 51/60 (85.0%) for controls, a gap of −13.3 points [−27.6, +1.5]. The interval includes zero, so the frozen rule returns *no detectable usability cost*; the point estimate lies below the −10 deployment line, so that clause is unresolved, not passed, and the audit can detect a gap of about 18 points near the observed control rate, so this is not evidence of no cost. Only this usable endpoint was the frozen decision rule. Two stricter secondary endpoints, defined before the returns and uncorrected for multiplicity, do show a cost against never-failed controls: meaning preserved 79.7% against 93.2% (−13.6 points [−26.2, −1.0]) and usable *and* preserving 61.7% against 81.7% (−20.0 points [−34.8, −3.9]); the looser usable-or-minor endpoint, reported for completeness, is 90.0% against 93.3% (−3.3 points [−14.2, +7.3]). The cost is uneven by model (descriptive, an unbalanced list): the smallest model's repairs were usable in 6 of 19 texts, the API model's in 12 of 12. Against its own failed original (a paired contrast on 20 pairs), a repair keeps the meaning in 6 of 20 pairs and the original in none (+30.0 points [9.1, 46.8], sign test $p = 0.03$; the other 14 pairs are unchanged on meaning), and it removes invented stance (a meaning, certainty, politeness or speaker the input did not give) in 10 of 19 pairs and adds it in none (−52.6 points [−69.1, −26.4], $p = 0.002$). Repairs keep meaning more often than their originals but are not more usable: paired usable −5.0 points [−20.2, +10.8], usable and preserving +10.0 [−9.4, +29.1], $p = 0.62$.

	Development 58 families, 6 dimensions, 1,392 cue targets	**Confirmation (frozen)** 78 families, speech act, 1,872 targets
Benefit (% of targets)		
Strict-check pass: first pass / strip-only / guard	94.25 / 96.70 / 98.78	74.47 / 81.73 / 87.87
Pass (joint): same order	94.68 / 96.77 / 98.85	78.79 / 82.26 / 88.62
Guard − first pass, points	+4.53 strict-check	**+9.83** [8.39, 11.32] pass; +13.41 [11.38, 15.44] strict-check
Strict-check failures repaired (repair rate)	63/80 (78.8%)	251/478 (52.51%)
Free baseline: strip-only, zero calls (+2.44 strict-check development, +3.47 [2.78, 4.22] pass confirmation)		
Guard − strip-only, points	**+2.08** [1.15, 3.16] strict-check	+6.36 [5.29, 7.48] pass
Free baseline 2: resample without feedback (G1r), T=0.7, same retry cap; G1t is the guard rerun at T=0.7		
Pass: resample G1r / guard G1t (T=0.7)	95.6 / 98.6	82.4 / 89.4
Repair rate: resample G1r	15/80 (18.8%)	87/478 (18.2%)
Guard (T=0) − G1r, points (pass)	**+3.23** [1.87, 4.81]	**+6.20** [4.75, 7.64]
G1t − G1r, matched decoding (pass)	+2.95 [1.65, 4.38]	+6.94 [5.34, 8.60]
What remains (offline failure indicators)		
Residual strict-check failures	17/1,392 (1.2%)	227/1,872 (12.13%)
Already-correct text changed	14/1,312 cue; 49/2,154 all inputs	26/1,394 (1.87%)
Strict-check pass → failure	0/1,312	0/1,394
Cost and verdict		
Audit vs controls (points; usable / preserving / both)	—	−13.3 [−27.6, +1.5] / −13.6 [−26.2, −1.0] / −20.0 [−34.8, −3.9]
Extra calls per input	0.104 (1,856 inputs)	0.41 (1,848 inputs)
Pass line	passed	passed on the three scored criteria; usability clause unresolved (§6.2)

Visible input	First pass	Guard output
Development set, repaired (Gemma, one cue; 1st by hash order; 63/80 scored failures repaired)		
`[PLAIN] let finish sentence`	"Please let me finish my sentence."	"Let me finish the sentence."
Development set, failure that remains (Llama, all cues; 1st by hash order; 17/80 failures remain, 14/17 still flagged; the source asks whether the listener is charging it)		
`[?] [YOU] charge wheelchair`	"Can you charge my wheelchair?"	"Can you charge my wheelchair?"

Table 2: The guard as one display. Left: benefit, the free baseline and what remains, on two *separate* populations that are never pooled; scored partial predicates, not usability. *Targets* are scored (item, model, input condition) outcomes; *inputs* are the distinct requests the guard processed. Development: 1,392 cue targets, plus 464 requests with no supplied cue (no strict target), give 1,856 inputs. Confirmation: 1,872 targets share 1,848 distinct requests. Pass means joint pass; strict-check rows are labelled. Differences use unrounded rates, so a difference of printed levels can differ by 0.1. Right: stored examples, verbatim, by the rule in each row; not typical. Brackets: 95% whole-family bootstrap intervals for paired differences. Repair rates condition on initial strict-check failures; already-correct and pass-to-failure rows on initial strict-check passes.

6.3 Two more ledger rows: the menu and the cue

The complete menu raises coverage of both polarity or certainty readings (78/80 sets against 15/80; +78.75 points; passes its line), but on other dimensions it removes the polite option (26/40 to 12/40 politeness sets, 14/31 to 6/31 speech-act sets): for `knock come`, "Please knock before you come in." became "Don't knock before you come in." (Figure 1). The cue keeps the critical value in 94.8% of development

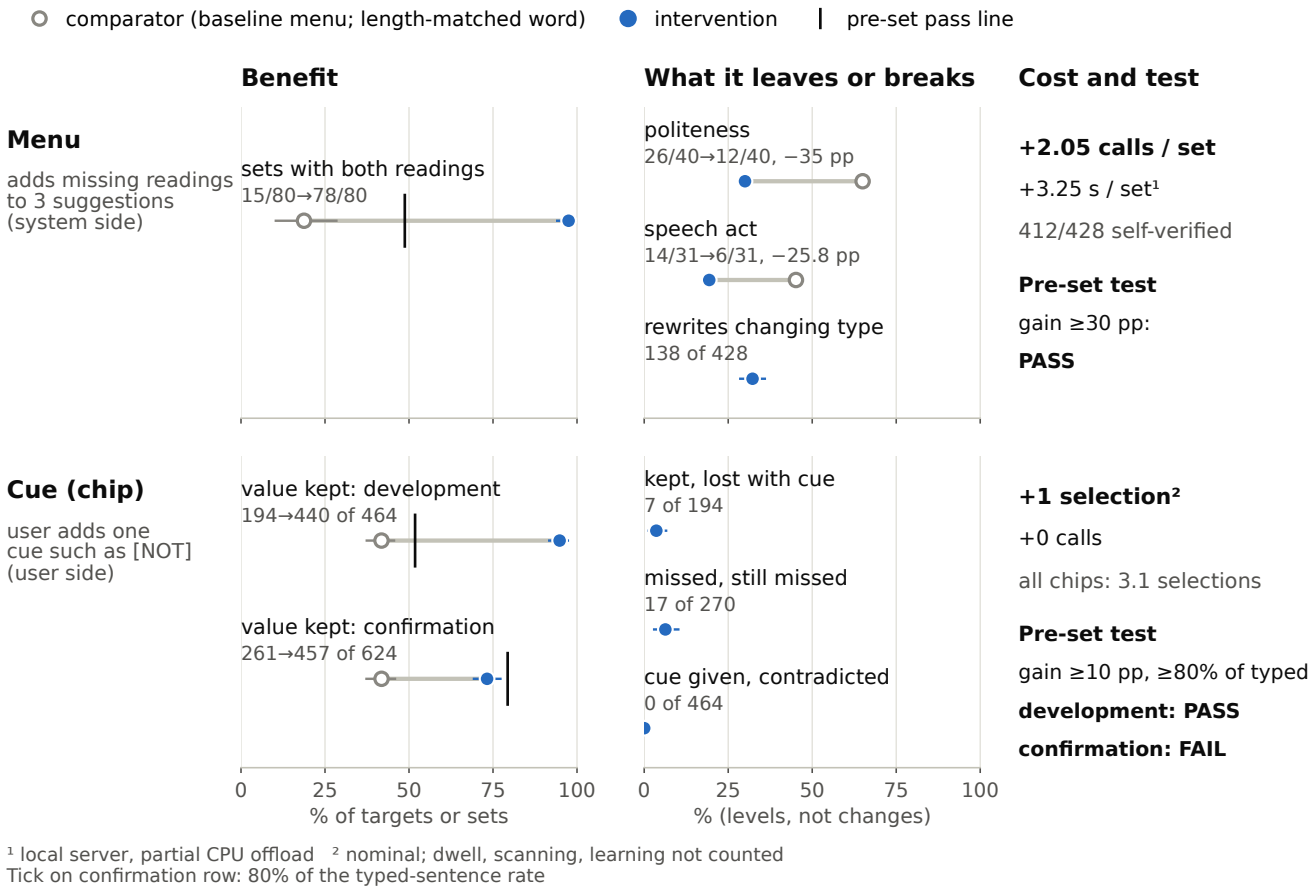


Figure 2: Two add-ons help in one place and cost something elsewhere (machine-extracted labels; four models). Open circle: comparator; filled: intervention; bars: 95% cluster-bootstrap intervals; black tick: the pass line set in advance; pp: percentage points. Plotted values are percent *levels*; changes in points are printed. The menu's politeness and speech-act rows come from families different from its target gain (40 and 31 sets). The cue passes both clauses on the development set but *fails* its ≥80%-of-typed clause on the confirmation set (only its gain clause replicates). The guard is in Table 2.

targets against 41.8% (97.3% of typed input; passes both clauses), and its gain over the length-matched word replicates on 78 fresh speech-act families (31.4 points, or 18.6 [10.0, 26.9] if "Can you. . . ?" counts as a question), but one cue reaches 73.8% of typed input, below its 80% line, and question cues fail most visibly: `[?] parcel signature` became "Can you sign for this parcel?". The bare-keyword default is not a prompt artifact. On bare polarity inputs (4 models, both sources), affirmative output among committed answers was 99.5% with the cue legend, 98.4% with the legend removed and 99.9% when the prompt said markers may be missing and not to assume one.

6.4 Realistic-input checks

We ran the pipeline on 547 minimal pairs written by 20 non-author annotators and on keywords that 20 annotators typed for 600 AACText messages. On the human-written pairs the cue gain replicates on all 6 dimensions (35.3 to 82.7 points), and the three confirmation contrasts replicate in direction: speech-act cue +35.3 points, politeness addition +7.6, and a bare affirmative default of 99.8% [99.3, 100.0]. Two controls on these pairs show that the missing cue, not the chip, is the lever. An irrelevant chip from another dimension adds little: the cue chip beats it by 30.0 to 80.9 points by dimension. The same cue as a plain word (`not`, `please`) is within 2.1 points of the chip on 5 of 6 dimensions and beats it on speech act by 8.2 points. On Qwen2.5-14B and Mistral-Nemo-12B the speech-act gain is +35.1 and +37.3 points. With 6–7 writers per contrast we claim direction and consistency only: every writer's mean has the same sign, and the replication is exploratory. In the keyword study about one keyword set in five lost the negation (20.8% [16.9, 25.2]; 21.4% [15.4, 27.8] after resampling messages and reweighting to the sampling frame), on cue-enriched messages that carry a negation; a four-word compressor lost it 100.0%.

Defaults: a good guess with an asymmetric cost. When keywords carry no negation word, the models write an affirmative sentence for 99.9% of such inputs. That is the Bayes-preferred guess: 8.6% of AACText messages are negated, and among human keyword sets with no negation word the

source was negated only 2.7% [1.8, 3.7] of the time (2.0% from corpus prevalence), against 11.4% for the crude compressor; guessing affirmative is right 97.3% of the time. The harm is the asymmetric cost: about 24 [15, 33] of every 1,000 messages are silently reversed when the user types keywords unaided (a posterior-based estimate from the keyword study; Table 3 lists it with the two other estimates of the same quantity), against 0.8 spurious negations, and the guess breaks even only if a reversal costs under 36 times a spurious negation (the posterior odds, $(1-p)/p$ with $p = 2.7\%$, not the ratio of the rounded per-1,000 counts). For politeness the models match the preferred guess (6.7% against a posterior of 6.5%); for speech act they over-commit to "statement" (47.9% of sources were a question or request, models asserted one 28.2%). An accuracy metric rewards this behavior, so a ledger puts a default's cost beside its hit rate. The default holds on the 6 models from 5 families (at least 99.0% affirmative) and on a lite Gemini model (97.8% and 98.9% on the legend arms). Chen et al. (2023) reported that models write sentences wrongly positive about negative commonsense facts in keyword-to-sentence generation; their measurements are on negative commonsense facts, not AAC messages; we price the default against a human base rate on AAC-style messages and show it survives "do not assume" prompting (detector labels, one greedy draw per model).

6.5 Two ledger columns from people: writer intent and clinical weight

Writer intent (first test of H1). Each of 20 writers judged 10 outputs generated from their own pairs (200 in all), choosing the version they meant. A cue raises the share matching the intended version from 11.2% to 39.2%, +28.0 points [14.8, 42.0] (writer-clustered; keywords-only is a reference level, not a failure rate). Yet writers accepted 1 of 200 outputs as written, and the judge's critical-dimension label agrees with the writer's four-way label at only $\kappa = .24$. Keeping the critical commitment is necessary, not sufficient. Among the 130 outputs whose critical value the judge scored correct (126 with a decided answer; 4 unsure), writers rejected 77/126 as not the version meant. A post hoc decomposition (exploratory; classifier agreement with 40 hand-coded items $\kappa = .71$, optimistic because its rules followed those items) finds mainly compression loss and event/role misresolution among the judge-correct outputs writers did not accept: 98% of the writers' sentences contain words no keyword supplied, 49% of the outputs change who does what, and added content is the primary reason in 0 cases but is present in 22% of these outputs (the judge's added-content flag: 22%). Two AAC-user design consultants independently asked that relations (who does what) be preserved.

Clinical weight. Two speech-language pathologists (expert review of study materials, not a population estimate) rated how serious each of seven kinds of silent meaning change is; both rated a dropped or added negation 5 of 5 and would block it. Weighting each judge-flagged failure by the severity of its type, the guard lowers the clinically weighted failure rate from 22.0% to 14.7% on the confirmation set, and a supplied cue lowers it on human-written pairs from 57.4% to 14.9% (chip) and 17.8% (word). On selected judge-failed items (a descriptive check, not comparable arms) they judged guard repairs acceptable to speak as they stand in 13/40 rater-judgments, against 0/20 for the originals of the same inputs; on these judge-selected items the SLPs accepted far fewer outputs than the machine judge did.

Estimator	Unit resampled	Silent reversals per 1,000
Posterior (Sec. 6.4)	messages, reweighted to the sampling frame	24 [15, 33]
Model-visible	exposure units	23.9 [16.0, 32.5]
Bare-arm regeneration	same items, regenerated bare (point rate)	23.8

Table 3: Three estimators of one quantity, not independent replications. The keyword-loss share (20.8% of keyword sets lost the negation) is a different quantity.

6.6 Frontier limits: the cue's measured price, and what asking and uncertainty do not do

These results are provisional: stored outputs, machine labels resolved by a blind arbiter, the three local primary models (Gemma-3-4B, Llama-3.1-8B, Qwen2.5-7B), 332 exposure units. Always committing leaves about 23.9 silent negation reversals per 1,000 messages (Table 3); the abstract and the cue comparison below use the bare-arm regeneration rate, 23.8. Supplying the cue on every message, measured by regeneration on the same items, cuts polarity reversals per 1,000 from 23.8 to 1.6 (chip) and 2.6 (word), and politeness reversals from 51.8 to 0.7 and 1.2; speech-act restoration is 75.6% with a chip and 90.4% with chip plus guard. Asking only on some messages shows no detectable benefit over gated-random asking: none of the 4 ask-policy contrasts is significant after Holm correction (minimum detectable difference 8.5 per 1,000), and a CLAM-style clarification trigger leaves 21.2 reversals per 1,000 against 21.3 for gated random at a 10% budget (Holm $p = 1.00$); power is limited, not equivalence. Model uncertainty does not flag them either: with 16 samples, Qwen and Llama write a negated sentence for 0.4% and 1.9% of no-cue inputs against 2.6% expected from the human base rate, and the sampled share ranks at-risk inputs at AUROC .56 against .75 for a lexical model. The cue has to come from the user.

7 From Ledger to a Pre-Specified Human-Effect Study

Each prediction names the study that tests it and what would demote it; H1 and H2 were tested on data already seen (post hoc tests of earlier predictions), and the H3 design is pre-specified in version control.

Predictions. H1 (writer intent). Among outputs the scorer counts as preserving a writer's critical value, writers report that the output "said what I meant" more often than among outputs it counts as failing; a gap whose interval

includes zero demotes the scored endpoint. The first test is the writer-intent check of Section 6.5. This first, post hoc test is a consistency check, close to definitional: writers said the output was the version they meant for 49/126 outputs whose critical value the judge scored correct, against 0/69 of those it scored wrong (Section 6.5). It is necessary, not sufficient: 61.1% of the judge-correct outputs were still not the version meant. **H2 (repair usability).** A scored repair is usable at no less than 10 points below a never-failed control; the 120-text audit tested this under the frozen rule of Section 6.2: no detectable cost; deployment clause unresolved; stricter endpoints show a cost, so H2 is unresolved. **H3 (collateral).** Offline changed-correct and coverage loss predict a later drop in users' "said what I meant" rate; if not, evaluation of this kind should be demoted. H1 and H2 were testable with writers and raters; H3 needs AAC users, and no value in its design is a result.

Design implications from two AAC-user consultants. Two paid AAC-user design consultants rated our five requirements (keep a button's meaning; do not change typed text; do not quietly guess; a new button must not break what was right; an honest test screen) important, and each asked for a sixth (we report requirement changes only): **R6, a draft is never spoken without the person's approval**, with the original words kept beside the draft and speaking a separate deliberate act. They also asked for an exact-copy default for typed sentences, a visible scope for NOT, and showing where two drafts differ. R6 makes flag-and-confirm the default design, as writers' acceptance of 1 of 200 outputs as written implies.

Study for H3: a micro-randomized trial. *Estimand.* For adult AAC users who compose with LLM keyword expansion, the average causal excursion effect (Boruvka et al. 2018) on a proximal outcome of delivering the regenerated sentence versus the original, at decision points where the monitor flags a conflict. *Arms and unit.* Randomization is per flagged message within user (Klasnja et al. 2015), with fixed probabilities, among three arms: silent regeneration, flag-and-confirm (the user sees the flag and chooses), and the original. Cues can only be offered (an encouragement design), so they enter as a between-user factor. *Outcomes.* Proximal: sent without edit, one-tap "said what I meant" on a random subset of sends, time to send. Longitudinal: weekly rates of both, abandonment, and reliance (whether acceptance of flagged text rises over weeks), with week as a moderator. *Identifying assumptions.* Positivity (every flagged decision point has a nonzero chance of each arm, by design), consistency (the delivered sentence is the arm) and no interference across a user's decision points beyond what we model; carryover is the main threat, because acceptance may rise with reliance, so we target the *excursion* effect at a decision point (deviate once, then resume usual practice) and track week and reliance as moderators. *Sample size* (illustrative two-contrast calculation on typed assumptions to be replaced by pilot estimates). The monitor flagged about 8.5% of constructed inputs (real-use rate unknown), so 40 randomized decision points per user need roughly 473 messages; with the outcome on a random 50% of sends, a standardized effect of 0.2, within-user correlation 0.3 and 80% power at the 5% level, about 132 users are needed (125 if every send were rated). *Analysis:* weighted, centered least squares with user-clustered errors. *Power of the writer-intent check.* Effective n 71 at an assumed within-writer correlation of 0.2 detects about 23 points in the worst case.

8 Limitations

Human outcomes. Results are changes in stored text; communication success, agency, effort and trust are untested (Section 7). *Repair audit.* The audit can detect a gap of about 18 points; its usable interval includes zero and a sizeable cost, and its raters are non-author peers, not AAC users. *Machine labels.* Endpoints use an LLM extractor validated by two annotators on untreated outputs only. *Baselines and scope.* Strip-only covers one failure type; the confirmation is qualified by early cached requests and post-outcome arbitration; the menu was not run on the confirmation set; the human-written pairs have few writers. *Constructed items, fixed models.* Open models are at most 14B and 4-bit; AACText is crowd-written fiction and word input is the only modality. *Community involvement.* Two paid AAC-user design consultants and two speech-language pathologists reviewed requirements and study materials (n=2 each; expert review, not a population estimate); the requirements remain our proposals, with no claim of benefit to AAC users; we draw on AASPIRE (Nicolaidis et al. 2019), the double-empathy (Milton 2012) and heterogeneity (Zisk and Dalton 2019) cautions. *Ethics.* No data were collected from AAC users. All annotators were adult non-authors recruited from the authors' university peers, identified only by an ID; each consented and was reimbursed for their time. They labelled model outputs and audited items under a fixed codebook, blinded to model, condition and source; some also chose keywords for public AACText messages or wrote new sentence pairs, released under CC BY 4.0 with recorded consent. No personal or health data were collected, and we did not seek an institutional review determination.

9 Conclusion

On fresh confirmation families the guard raises the pass rate by +9.83 points, repairs 52.51% of strict-check failures with 0 pass-to-fail changes, and beats resampling at matched temperature (+6.94 points). In the human audit its repairs keep the meaning and remove invented stance relative to their failed originals (a paired contrast on 20 pairs); against never-failed controls the frozen rule finds no detectable cost, the deployment clause is unresolved and the stricter endpoints show a cost. A supplied cue removes nearly all silent reversals on the same items. The ledger records what remains (deletion alone earns +3.47 of the +9.83 points; writers accept 1 of 200 outputs as written), and the pre-specified study, with R6 as a requirement, turns these into predictions that AAC users can test. Scripts that rebuild every number accompany this submission.

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